Let $q(\lambda)$ be the max among: # enc queues and length of all queues. Wlog assume that $t(\lambda) \le t^*(\lambda) \le q(\lambda)$.

$$r, r+1, r+2, \ldots, r+q-1$$

$$r^*, r^*+1, r^*+2, \ldots, r^*+q-1$$

query $i \in [q]$

$$\Pr[\text{OVERLAP}] \le \sum_{i=1}^{q} \Pr[\text{OVERLAP}_i]$$

$$\text{OVERLAP}_i \iff r^* - q + 1 \le r \le r^* + q - 1$$

$$\Pr[\text{OVERLAP}_i] \le (2q-1)/2^n = \text{negl}(\lambda)$$

Made with Goodnotes

Exercises

*) Prove or refute: Any SKE Π is perfectly secret iff: $\forall$ M over $\mathcal{M}$, $\forall c_0, c_1 \in \mathcal{C}$

$$Pr[C = c_0] = Pr[C = c_1].$$

$$C = Enc(K, M)$$

$K \equiv$ uniform

Let M be uniform over $\mathcal{M}$ and let

$$Enc(K, M) = M = C$$

Made with Goodnotes

Thus does not work because the def. says

$\forall$ M over $\mathcal{M}$.

We need to find (Enc, Dec) PERFECTLY

SECRET but s.t. C is NOT UNIFORM.

E.g.: To $\mathcal{M}$ $c = k \oplus m$ (OTP)

Let $c' = .0 \,||\, k \oplus m$

$$\mathcal{M} = \{0,1\}^n = \mathcal{K} = \mathcal{C}$$

$$\mathcal{C}' = \{0,1\}^{n+1}$$

Made with Goodnotes

Take c_0 to be any string starting with a

and c_1 " " " " " " 1

$$\Pr[C = c_0] \neq \Pr[C = c_1]$$

On the other hand the SKE is still PERFECTLY SECRET.

*) Let $G: \{0,1\}^n \to \{0,1\}^{2n}$ be a PRG.

Show that G is also a OWF.

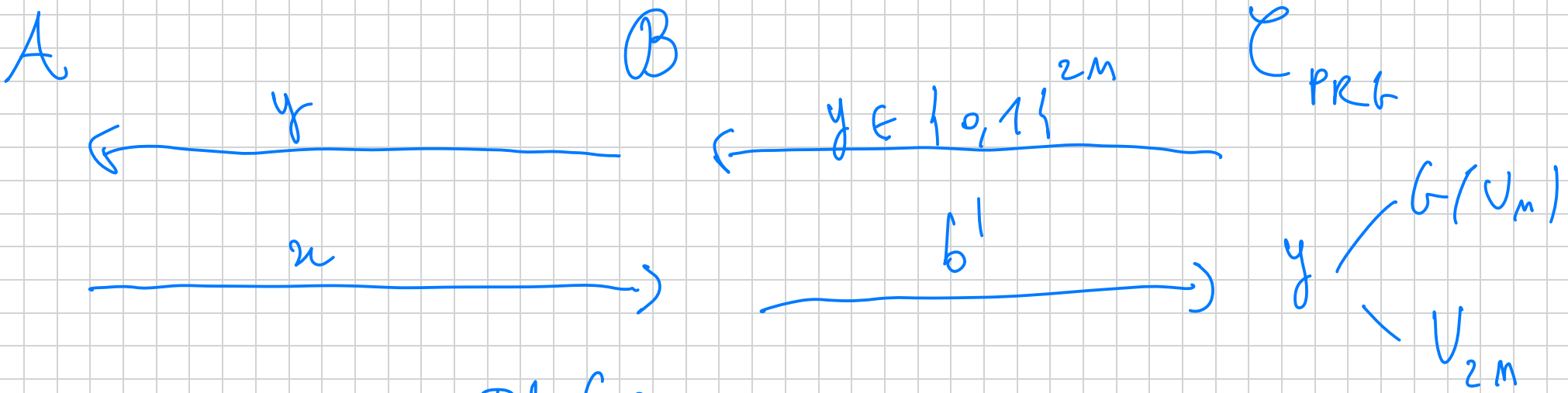
Assume not: $\exists$ PPT A s.t.

$$\Pr[G(A(y)) = y : y \equiv G(U_n)]$$

$$= \varepsilon(n) \geq 1/\text{poly}(n)$$

Made with Goodnotes

Then, we build PPT B against PRG.



If $G(x) = y$

$b' = 1$

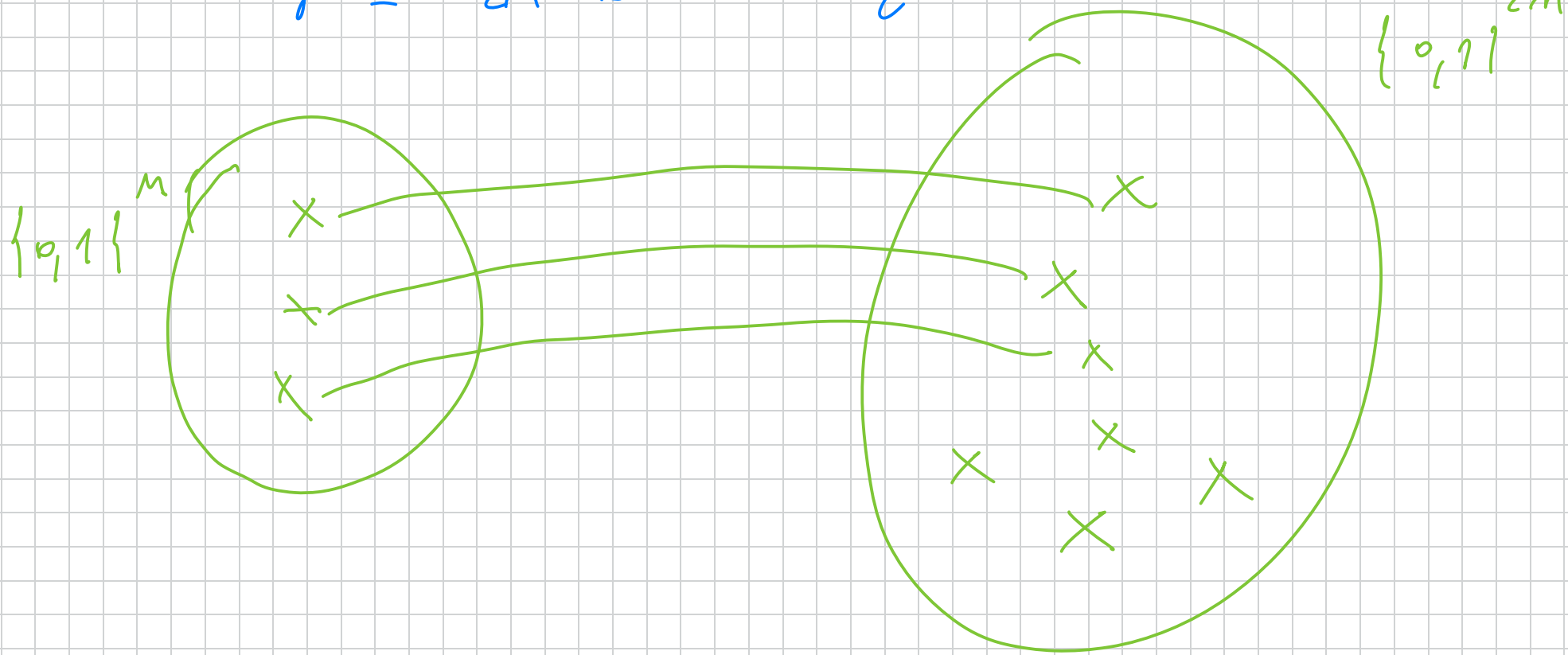
Else $b' = 0$ (Else $b' \leftarrow \{0,1\}$)

$$Pr[b' = 1 : y \equiv G(U_m)] = Pr[G(A(y)) = y : \overset{y \equiv}{G(U_m)}]$$

$$= \varepsilon(n)$$

Made with Goodnotes

$$Pr[b' = 1 : y \in U_{2n}] \leq \frac{2^n}{2^{2n}} = 2^{-n}$$



$$Adv(B) \geq \varepsilon(n) - 2^{-n} = 1/poly(\lambda).$$

$$= \left| Pr[b' = 1 : y \in G(U_n)] - Pr[b' = 1 : y \in U_{2n}] \right|$$

Made with Goodnotes

Difficult exercise: Same as above but

$$G : \{0,1\}^n \to \{0,1\}^{n+1}$$

*) For each of the following PRFs say if it is secure or not:

(i) $F_k(x) = G'(k) \oplus x$ where

$$G : \{0,1\}^\lambda \to \{0,1\}^{\lambda+l} \quad \text{and} \quad x \in \{0,1\}^\lambda$$

and $G'(k) = G(k)|_\lambda$

(ii) $F_k(x) = F_x(k)$

(iii) $F'_k(x) = F_k(0 \| x) \| F_k(x \| 1)$

Made with Goodnotes

$$x \in \{0,1\}^{n-1}$$

$$F_k : \{0,1\}^{\wedge} \to \{0,1\}^{n}$$

(NV) $F_k'(x) = F_k(0 \| x) \| F_k(1 \| x)$

$y_1 =$

(i) NO. A queries x_1 and gets $G'(k) \oplus x_1$

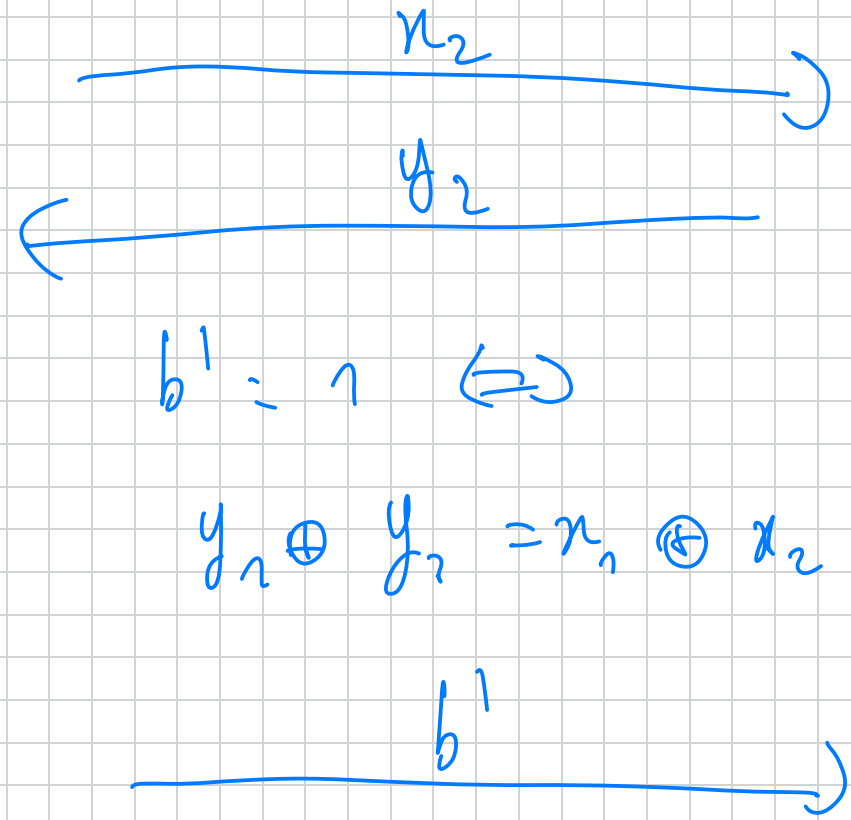
x_2 " " $G'(k) \oplus x_2$

$= y_2$

$$y_1 \oplus y_2 = x_1 \oplus x_2 .$$



Made with Goodnotes



(NN) No. We need to find $\sqrt{F}^{BAD}$ s.t. PRF

$F'_K(x) = F^{BAD}_x(K)$ not PRF

What about $\sqrt{F}$:

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$$F_K^{BAD}(x) = \begin{cases} 0^n & \text{if } K = 0^n \\ F_K(x) & \text{for PRF } F \end{cases}$$

F^{BAD} still a PRF as $K = 0^n$ happens

w.p. $\leq 2^{-n}$

But $F_x^{BAD}(K)$ is A can choose $x = 0^n$

and output $b' = 1 \iff y = 0^n$.

(NNN) Take $x_1 = 0^{n-1}$ and $x_2 = 0^{n-2}11$

$$F_K'(x_1) = F_K(0^n) \,||\, F_K(0^{n-1}11)$$

$$F_k'(x_2) = F_k(0^{n-1}\|1) \| F_k(0^{n-2}\|1^2)$$

(NO) YES.

*) Prove that no SKE (Enc, Dec) can be CPA secure against UNBOUNDED ATTACKS.

Made with Goodnotes